MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 5639

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Penetode, subminiature, video amplifier

Outline - - - 3-3(EIA)
Base - - - E8-10
Envelope - - - T3
Cathode - - - Coated unipotential

Base connections:

Pin No. - - -	1	2	3	4	5	6	7	8
Element - - -	g1	k,g3	h	k,g3	a	h	g2	k,g3

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Ehk	Rk	Rg1	Ik	Pp	Pg2	TE	Alt
Unit:	V	Vdc	Vdc	Vdc	v	Ohms	Meg	mAdc	W	W	°C	ft
Maximum:	6.6	165	0	155	200	---	0.5	40	3.5	1.0	220	Note 1
Minimum:	6.0	---	-55	---	---	---	---	---	---	---	---	---
TEST CONDITIONS:	6.3	150	0	100	0	100	---	---	---	---	---	---

GENERAL:

Qualification - Required

Reliable tube

5639

FSC 5960

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Method	Requirement or test	Notes	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits		Unit
							Min	Max	
	Quality conformance inspection, part 1								
1256	Electrode current (1) (anode)			0.65	II	Ib	14	28	mAdc
1256	Electrode current (2) (anode)		Ec1 = -14 Vdc; Rk = 0	0.65	II	Ib	---	75	µAdc
1266	Total grid current	2	Rg1 = 1.0 Meg	0.65	II	Ic1	0	-1.0	µAdc
1301	Heater current			0.65	II	If	420	480	mA
1306	Transconductance (1)			0.65	II	Sm	7,500	10,500	µmhos
1336	Heater-cathode leakage			0.65	II	Ihk	---	15	µAdc
1201	Short and discontinuity detection			0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1211	Insulation of electrodes			2.5	I	R	---	---	---
1031	High-frequency vibration		Rp = 2,000 ohms	2.5	I	Ep	---	100	mVac
1246	Audio frequency noise		Esig = 200 mVac; Ecc2 = 100 Vdc; Ec1 = -2.5 Vdc; Rg1 = 0.5 Meg; Rg2 = 0.01 Meg; Rp = 2,000 ohms; Rk = 0; Cg2 = 4 µF	2.5	I	EB	---	17	vu
1256	Electrode current (screen grid)			2.5	I	Ic2	2.0	6.0	mAdc
1266	Grid emission	3	Ef = 7.5 V; Ec1 = -14 Vdc; Rg1 = 1.0 Meg; Rk = 0	2.5	I	Ic1	0	-2.0	µAdc
1306	Transconductance (2)		Ef = 5.7 V	2.5	I	ΔSm Ef	---	10	%
1311	Anode resistance			---	---	Rp	0.033	---	Meg
1331	Direct-interelectrode capacitance		Shield No. 318	6.5	Code F	Cglp Cin Cout	--- 8.0 7.0	0.13 10 9.0	pF pF pF
1341	Power output		Esig = 2.0 Vac Rp = 9,000 ohms	2.5	I	Po	0.75	---	W
1116	Lead fatigue			2.5	Code F	---	---	---	---
2126	Envelope strain			2.5	I	---	---	---	---

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Method	Requirement or test	Notes	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 2 - Continued								
1041	Shock		450 G; Ehk = +100 Vdc; Rg1 = 0.1 Meg	6.5	See note 4	---	---	---	---
1031	Vibration-fatigue			6.5	See note 4	---	---	---	---
---	Shock and vibration-fatigue test end points:								
1031	High-frequency vibration			---	---	Ep	---	350	mVac
1336	Heater-cathode leakage			---	---	Ihk	---	40	µAdc
1306	Change in transconductance (1) of individual tubes			---	---	ΔSm t	---	20	%
1105	Permanence of marking			---	---	---	---	---	---

Method	Requirement or test	Notes	Conditions	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 3						
1506	Heater-cycling life		Ef = 7.0 V; 1 min on, 4 min off; Ehk = 140 Vac; Ec1 = Ec2 = Eb = 0	---	---	---	---
1516	Stability life		Ehk = +200 Vdc; Rg1 = 0.5 Meg; TA = room	---	---	---	---
---	Stability life-test end point:						
1306	Change in transconductance (1) of individual tubes			ΔSm t	---	10	%
1501	Intermittent life	6	Ehk = +200 Vdc; Rg1 = 0.5 Meg; TA = room; Group E	---	---	---	---
---	Intermittent life test end point (500 hours):	5					
1211	Insulation of electrodes			R	50	---	Meg
---	Intermittent life test end point (1,000 hours):	5					
1211	Insulation of electrodes			R	25	---	Meg
1501	Intermittent life (high temperature)	6	Ehk = +200 Vdc; Rg1 = 0.5 Meg; TA = room; TE = +220°C(min); Group E	---	---	---	---

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Method	Requirement or test	Notes	Conditions	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 3 - Continued						
---	Intermittent life test end points (high temperature) (500 hours):	5					
1266	Total grid current			Ic1	0	-2.0	µAdc
1301	Heater current			If	414	492	mA
1306	Change in transconductance (1) of individual tubes			ΔSm t	---	20	%
1306	Transconductance (2)			ΔSm Ef	---	15	%
1336	Heater-cathode leakage			Ihk	---	60	µAdc
1211	Insulation of electrodes			R	50	---	Meg
1306	Transconductance (1) average change			Avg ΔSm t	---	15	%
---	Intermittent life test end points (high-temperture) (1,000 hours):	5					
1266	Total grid current			Ic1	0	-2.0	µAdc
1301	Heater current			If	414	492	mA
1306	Change in transconductance (1) of individual tubes			ΔSm t	---	25	%
1306	Transconductance (2)			ΔSm Ef	---	20	%
1336	Heater-cathode leakage			Ihk	---	60	µAdc
1211	Insulation of electrodes			R	25	---	Meg

NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage.
2. This test shall be performed at the conclusion of the holding period.
3. Prior to this test, tubes shall be preheated 5 minutes at conditions indicated below. Test within 3 seconds after preheating. The 3-minute test is not permitted. Grid emission shall be the last test performed on the sample selected for the grid-emission test.

Ef	Ec1	EC2	Ec3	Eb	Rk	Rg1
V	Vdc	Vdc	Vdc	Vdc	Ohms	Meg
7.5	0	100	0	150	100	0.5

4. This test shall be conducted on the initial lot and thereafter on a lot approximately every 12 months. When one lot has passed, the 12-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code letter E, shall apply.
5. The allowable defectives per characteristic shall not be greater than 1 for the first sample and shall not be greater than 3 for combined samples.
6. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (method 1226), will be satisfied if a TUT having bogey Ib (±5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.
7. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

User activities:
Army - AR
Navy - AS, OS, MC, CG
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DLA - ES

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